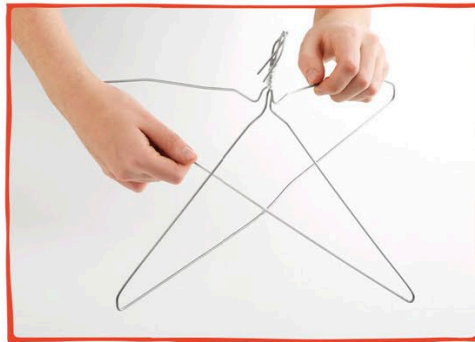


TAKE IT FURTHER

A great way to display your planets is to make a mobile to hang in your bedroom. You'll need to make a model of the sun. In real life, the sun's diameter is more than a hundred times that of Earth: for the mobile make it with fifteen pieces of paper and yellow rubber bands.



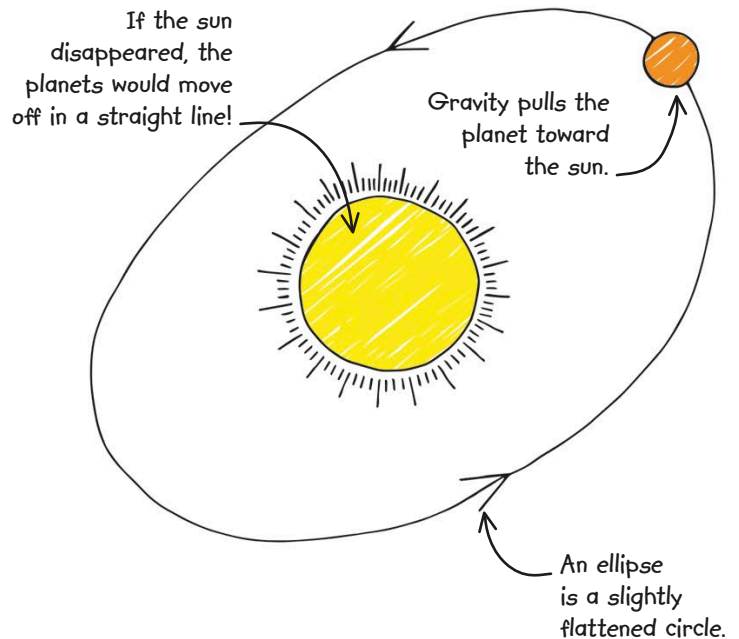
1 To make the mobile, cross over two wire coat hangers and secure them top and bottom with fishing wire or tape. Tie about 12in (30cm) of fishing wire around your sun and planets, so they are secure.



2 Tie the free end of each length of fishing wire around the coat hangers, so the sun and planets all hang down at different heights. Make sure you put the sun in the center. Ask an adult to help you hang your mobile.

HOW IT WORKS

The planets are huge objects hurtling through space. Mercury travels the fastest, with an average speed of more than 105,000 miles (170,000km) per hour! Despite their high speeds, the planets don't head straight off into space. Instead, they follow orbits, pulled toward the sun by the force of gravity—the same force that makes you drop back to the ground when you jump in the air. All space bodies, including planets, move in an elliptical orbit. Gravity keeps the moon and satellites in orbit around Earth, too.



REAL WORLD SCIENCE

DISTANCE FROM THE SUN

The distances between the sun and planets are enormous. For scale, if you were to place your model Earth at the correct distance from the flashlight, they would be 820ft (250 meters) apart! The distances between planets increases the farther out you go.

